THE GEOMETRICALLY IRREDUCIBLE FIBRE LOCUS OF A MORPHISM IS STABLE UNDER ARBITRARY BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 7431-7441 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 7443-7450, the complete author proof (6 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The Varieties lemma that geometric irreducibility of a scheme over a field is preserved and reflected by arbitrary field extension; cited input with the fibre identification written in the retained span.

Source reference. Original lemma 054P: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7450–7450. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Let $g : Y' \to Y$ be any morphism, and denote $f' : X' \to Y'$ the base change of f. Then*

$$\{y' \in Y' \mid X'_{y'} \text{ is geometrically irreducible}\}$$
$$= g^{-1}(\{y \in Y \mid X_y \text{ is geometrically irreducible}\}).$$

Proof. This comes down to the statement that for $y' \in Y'$ with image $y \in Y$ the fibre $X'_{y'} = X_y \times_y y'$ is geometrically irreducible over $\kappa(y')$ if and only if X_y is geometrically irreducible over $\kappa(y)$. This follows from Varieties, Lemma 054P. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.